FEDERAL STANDARD STOCK CATALOG

Section IV

(Part 5)

FEDERAL SPECIFICATION

FOR

PAULINS AND COVERS; DUCK (TARPAULINS)

This specification was approved on the above date by the Director of Procurement, for the use of all departments and establishments of the Government, and shall become effective not later than September 1, 1940. It may be put into effect, however, at any earlier date after promulgation.

A. APPLICABLE SPECIFICATIONS.

A–1. The following Federal Specifications, of the issue in effect on date of invitation for bids, shall form a part of this specification:

T–R–571—Rope; Cotton.
T–R–601—Rope; Manila.
V–T–276—Thread; Cotton.
CCC–D–746—Duck, Cotton; Fire, Water, and Weather Resistant.
CCC–D–771—Duck; Cotton, Plied-Yarns (Army, Numbered, and Tent-Duck).
CCC–T–191—Textiles; General Specifications, Test Methods.
DDD–S–751—Stitches; Seams; and Stitching.

B. TYPES, GRADE, AND SIZES.

B–1. *Types.*—Paulins (tarpaulins) and covers shall be of the following types as specified in the invitation for bids:

Type I.—Fire, Water, and Weather Resistant.
Type II.—Water Resistant.

B–2. *Grade.*—The grade shall be "Firsts."

B–3. *Sizes.*—Sizes shall be as specified in the invitation for bids. (See pars. C–1a and I–2.)

C. MATERIAL AND WORKMANSHIP.

C–1. *Material.*—Shall be suitable for the purpose and conform to the following:

C–1a. *Duck.*—Unless otherwise specified, the width of the duck shall be 26 inches, 30 inches, 36 inches, 42 inches, or 48 inches, and conform to the following:

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C–1a (1). *Type I.*—Shall conform to the requirements for Type I of Federal Specification CCC–D–746, and shall be of the number and texture specified in the invitation for bids. (See par. I–3.)

C1a (2). *Type II.*—Shall conform to the requirements for type I of Federal Specification CCC–D–771, and shall be of the number and texture specified in the invitation for bids. (See par. I–3.) The duck to be made highly water resistant with a compound evenly applied in such a manner as to completely penetrate the duck and not merely coat the two sides of the fabric. The treatment shall leave no material on the fabric which will be injurious to the skin in handling the treated material, or to wood or metal with which it may be in contact. The weight of the treated duck shall not exceed the standard weight of the untreated (see Federal Specification CCC–D–771) by more than 35 percent for No. 4 duck, 45 percent for No. 8, 55 percent for No. 12, and others in proportion. No fillers or loading materials of any kind shall be used. When tested according to paragraph F–3, the treated duck, both before and after aging, shall withstand a hydrostatic head as specified therein.

C–1b. *Rope.*—Shall be cotton or manila as specified in the invitation for bids and shall conform to the following:

C–1b (1). *Cotton.*—⅜ inch diameter; Federal Specification T–R–571.

C–1b (2). *Manila.*—⅜ inch diameter; Federal Specification T–R–601.

C–1c. *Grommets.*—Shall be brass grommets, or hand-worked, or machine sewn ring grommets with or without brass ferrules, as specified in the invitation for bids, and conform to the following:

C–1c (1). *Grommets and washers.*—Shall be rolled rim brass, No. 5 spur grommets and toothed washers.

C–1c (2). *Rings.*—Shall be malleable iron, galvanized, ⅛ inch round by ¾ inch I. D.

C–1c (3). *Ferrules.*—Shall be of brass of a suitable weight and of a size to fit snugly into the sewn grommet.

C–1d. *Thread.*—Shall conform to the requirements of type IIIA of Federal Specification V–T–276. For sizes see table I.

C–2. *Workmanship.*—The finished paulins and covers shall be clean, well made, and free from any defects which may affect appearance or serviceability.

D. GENERAL REQUIREMENTS.

D–1. *Size.*—The size of the finished paulin or cover shall be as specified in the invitation for bids. (See par. I–2.)

D–1a. *Size tolerance.*—A tolerance for length and width of minus 1 percent and any reasonable plus will be permitted, unless specified otherwise in the invitation for bids.

D–2. *Stitches, seams and stitching.*—All seaming and stitching shall be done with stitch type 301 (lock stitch) or 401 (double chain locked stitch); seam type LSa–2 for joining seams, EFb–1 for hemming, and LSd–1 for seaming reinforcement patches, and shall conform to the requirements of Federal Specification DDD–S–751. The joining seams shall run lengthwise of the paulin or cover. Cross seaming is not permitted.

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D–2a. *Joining seam construction.*—Shall be as follows:

TABLE I

Duck	Seam overlap	Gage	Thread	Stitches per inch
2	1½ inches	1 or 1¼ inches	8/8	4 to 5
4	1½ inches	1 or 1¼ inches	8/8	4 to 5
6	1, 1⅛ or 1½ inches	¾, ⅞ or 1 inch	8/4	5 to 6
8	1, 1⅛ or 1½ inches	¾, ⅞ or 1 inch	8/4	5 to 6
10	1, 1⅛ or 1½ inches	¾, ⅞ or 1 inch	10/3	5 to 6
12	1, 1⅛ or 1½ inches	¾, ⅞ or 1 inch	10/3	5 to 6

D–3. *Construction.*—

D–3a. *Hems.*—All raw edges shall be hemmed. The material shall be turned under 2½ inches with the raw edge turned in and stitched; finished hem to measure 2 inches wide.

D–3b. *Grommet reinforcement patches.*—Triangular reinforcement patches of the same duck as the paulin or cover shall be used to reinforce all corners and grommets and be made by cutting an 8-inch square in half on the diagonal. The corner patches shall be seamed to the corners with the square corner of the patch and paulin or cover coinciding; the other grommet patches shall be seamed to the edges of paulin or cover with the long side of the patch and the edge of the paulin or cover coinciding. All raw edges of the patches shall be turned in ½ inch and seamed, except where the edge of patch is sewn in with the hem.

D–3c. *Grommets.*—Shall be placed in each corner, at both ends of all joining seams and evenly spaced on both sides approximately 3 feet apart. The grommets shall be centered in the reinforcement patches with their centers 1 inch from edge of paulin or cover.

D–3c (1). If brass grommets are used, they shall be securely attached with a toothed washer.

D–3c (2). If hand-worked ring grommets are used, they shall be hand worked with 4 folds of 5-ply cotton twine well waxed, using not less than 24 stitches per grommet with the ends made secure.

D–3c (3). If machine-worked ring grommets are used, they shall be sewn with ¾ cotton thread, well waxed, in such a manner as to well cover the ring, and shall have the ends of the twine made secure.

D–3c (4). If brass ferrules are used with the hand-worked or machine-sewn ring grommets, they shall be of suitable weight and design, and properly inserted into the sewn ring grommet so as to adequately protect the stitching.

D–3c (5). The holes punched in the duck to receive the grommets shall be 1/16 inch smaller than the outside diameter of the grommet barrel.

D–3d. *Ropes.*—Shall be not less than 5 feet long and inserted through grommet and made secure with an eye splice having a 4-inch eye and a 3-inch splice. The free end of rope shall be securely hand whipped, with 5-fold waxed cotton twine, with not less than 10 turns, or shall be provided with galvanized or copper wire clips, as specified in the invitation for bids.

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D–4. *Stamping.*—Unless otherwise specified, there shall be stamped in the corner of each cover, in indelible ink, in letters not less than ½ inch in height, the following:

Type.
Size.
Name of contractor.
Date and number of contract.
Specification number.
Stock number.
Name of contracting office.

E. DETAIL REQUIREMENTS.

E–1. See section D.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Inspection.*—Each paulin or cover shall be thoroughly inspected to insure that all requirements set forth in this specification are complied with.

F–2. *Testing.*—The methods of testing described in Federal Specification CCC–T–191, together with the methods of sampling and testing of the component items referred to in the specifications listed in section A, wherever applicable, shall be followed.

F–3. *Water resistance test for type II.*—A single thickness of the paulin or cover shall be clamped over a circular opening 4½ inches in diameter, plus or minus 1/32 inch, in the bottom of a reservoir. Water shall be introduced into the reservoir to a depth specified in table II, and this depth maintained for 10 minutes by adding more water as required. The water leaking through the specimen in the 10-minute period shall be collected and its volume measured. The test area of the duck is then aged by soaking in water at room temperature for 24 hours followed by creasing wet, drying in air at 65 percent, plus or minus 2 percent, relative humidity and 70° F., plus or minus 2° F, and creasing dry. The creasing is done by grasping the cloth on the two sides of the area to be tested and drawing it back and forth over the rounded edge of an iron bar about ⅛ of an inch thick. The cloth on each side of the bar is held nearly parallel in order to fold it sharply. The cloth is drawn back and forth 20 times with one face toward the bar and then 20 times with the other face toward the bar. The leakage of water through the aged sample is then determined as above.

TABLE II

Number of duck, commercial designation before treating	Depth of water in testing water permeability	Maximum allowable leakage before and after aging
	Inches	*Milliliters*
2–4	16	10
6–8	12	10
10–12	8	10

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G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging.*—Each paulin or cover shall be neatly folded and tightly rolled into a compact bundle and securely tied with suitable rope.

G–2. *Packing.*—Unless otherwise specified, the paulins or covers shall be packed in commercial containers so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate, to point of delivery.

G–3. *Marking.*—Unless otherwise specified, shipping containers shall be marked with the name of the material, the size, type, class, and the quantity contained therein, as defined by the contract or order number under which the shipment is made, the name of the contractor, and the number and date of the contract or order.

H. REQUIREMENTS APPLICABLE TO INDIVIDUAL DEPARTMENTS.

H–1. The following departmental specifications of the issue in effect on date of invitation for bids and the following departmental requirements shall form a part of this specification.

H–2. *Army.*—

H–2a. United States Army Specification 100–2, Standard Specifications for Marking Shipments. Quartermaster Corps drawing No. 5–10–2, Paulins.

H–2b. *Methods of sampling and inspection.*—

H–2b (1). *Contractor's inspection.*—The contractor shall thoroughly inspect the finished paulins or covers for compliance with this specification prior to delivery to the United States Government for final inspection.

H–2b (2). *Sampling.*—Samples of any materials, components, etc., not furnished by the United States Government, entering into the manufacture of the article covered herein, shall be selected from time to time by the Government inspector and carefully examined and tests made to determine if they are in accordance with this specification.

H–2b (3). *Inspection.*—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or material shall not relieve the contractor of responsibility for faulty workmanship or material which may be discovered at any time prior to final acceptance. Final inspection of the finished paulins or covers shall be made either at point of production or at point of delivery designated in the contract or purchase order of procuring agency. In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

H–2c. *Marking.*—Shipments shall be marked as specified in United States Army Specification 100–2.

H–2d. *Note.*—The use of this specification for the type of articles covered herein is mandatory on all procuring agencies of the Army.

H–3. *Navy Department.*—

H–3a. Navy Department General Specifications for Inspection of Material. Copies of this specification may be obtained, without cost, upon application to the Bureau of Supplies and Accounts, Navy Department, Washington, D. C.

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I. NOTES.

I–1. *Colors.*—The following are considered standard: Khaki, olive drab, and for Type II, natural (not colored or bleached). Type I is not available in natural color. Procuring agencies should specify desired color.

I–2. Wherever possible procuring agencies should specify a width which is in multiples or combinations of 36-inch, 42-inch, and 48-inch duck, making due allowance for seams.

I–3. The number and texture of the duck should be chosen with respect to the intended service. The following are recommended: No. 4 medium texture for heavy work including marine service, No. 8 hard texture for truck covers and for paulins to be used in construction work, No. 12 hard texture for services requiring light weight covers.

I–4. It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determne compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid samples should be definitely stated, the specification to apply in all other respects.

I–5. This specification covers only the types, classes, grades, sizes, etc., of the commodity as generally purchased by the Federal Government, and is not intended to include all of the types, etc., which are commercially available.

I–6. An index of Federal Specifications may be purchased as noted in the paragraph next below, price to be obtained from the Superintendent of Documents.

I–7. Copies of this specification, and of those referred to in paragraph A–1, may be obtained upon application, accompanied by money order, or coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington, D. C. Price 5 cents each, except DDD–S–751, price 25 cents.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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